

34.1 A 28nm 83.23TFLOPS/W POSIT-Based Compute-in-Memory Macro for High-Accuracy AI Applications

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Rapidly expanding artificial intelligence (AI) models, for complex AI tasks, drive high-energy efficiency and high-precision requirements for AI processors [1– 6]. Floating-point CIM (FP-CIM) is a promising technique to improve energy efficiency and maintain accuracy. However, FP-CIM with FP32/FP16/BF16 suffers from a performance bottleneck due to its large storage requirements and its considerable MAC power. The emerging POSIT data format, exploiting dynamic bit width that adapts to varied data distributions, can use a low bit width to achieve nearly the same training and inference accuracy as high bit width FP (POSIT8 $\approx$ FP16 and POSIT16 $\approx$ FP32) [6]. The POSIT data consists of 4 parts: sign (S), regime (R), exponent (E), and mantissa (M). It is defined as POSIT(n, es), where n is the total bit width and es is the E bit width. The R and M bit width varies dynamically, allowing a data range and precision trade-off at runtime. If R is r bits, then M is m bits, where $m = n - r - es - 1$. R is a unary (thermometer) code with successive 0s and 1s: eg. 11110 and 00001. The decimal value of R is the number of successive 1’s count minus constant value 1 for positive values, and it is successive 0’s count for negative values (R=3 for “11110”, R=-4 for “00001”). POSIT computing requires the decoding R to get the POSIT value, which is $(-1)^S \times (2^{es})^R \times 2^E \times 1.M$. With the dynamic data expression to adapt data distribution, POSIT8 can achieve comparable accuracy to BF16 for image classification on VIT-B with a $7.12\times$ energy reduction.

However, the dynamic property of POSIT introduces three challenges for applying POSIT to CIM: (1) the dynamic regime processing of POSIT computing introduces additional extract and codec logic, which requires $2.62\times$ higher power than fixed-exponent processing in conventional FP for image classification on VIT-B.; (2) the dynamic bit width mantissa of POSIT clashes with the structured CIM architecture, which is designed

to store weights with a fixed bit width, causing a 41.3% CIM underutilization; (3) during mantissa accumulation, 66.8% of dynamically aligned mantissa have no overlap bits, where the full adders perform $X + 0 = X$ operations. Thus, the logic toggles inside the adders are redundant, wasting 62.5% of energy.

This paper presents a digital POSIT CIM macro (DP-CIM) with three features to tackle the challenges caused by POSIT's dynamic property: (1) a bi-directional regime processing unit (BRPU) that replaces the complicated codec logic with shifting and concatenation logic by using the successive 0/1 coding property of the regime, thereby saving 40.3% of regime processing energy; (2) a critical-bit pre-compute-and-store (CPCS) CIM that utilizes the spare bits to facilitate the CIM array to perform dual-bit MAC in each cycle, improving CIM cell utilization by 63%; (3) a cyclically-alternating computing-scheduling unit (CASU) that uses bit wise or operations to replace bit wise addition if two mantissas have no overlap bits, saving 56.9% of energy for the adder tree.

[Figure 34.1.2](#) shows the overall architecture of the DP-CIM macro, consisting of BRPU, 16 CIM cores with CPCS CIM arrays, 16 adder trees with CASU, a 28kB global SRAM, an FP-to-POSIT encoder, a POSIT-to-FP decoder, and a top-controller. Each CIM core contains 16 CIM arrays. Each CIM array has an 8×48 CIM sub-array, where every 4b CIM cell is a CIM unit. When loading data to DP-CIM, the FP-to-POSIT encoder transfers the FP data to POSIT format for storage. The CPCS detects the weight mantissa bit width to pre-compute and store a logical value into the spare CIM cell. During computing, BRPU performs shift-OR-based regime processing. The CIM core, with CPCS, has three work modes, including 3b mode, 2-4b set-associate mode, and normal mode. The former two modes use the pre-computed and stored logical value to achieve inter-CIM dual-bit MAC. The CASU saves energy of the adder tree by substituting the addition to bit-wise or operation for no overlap mantissa. Besides, it changes the computing order of CPCS to increase the number of no-overlap mantissa.

[Figure 34.1.3](#) shows the BRPU, saving the regime processing energy. In each cycle, a fetch unit gets $2 \times 12b$ R_A and R_B for computing. Since the regime uses successive 1/0s count to represent value, the addition or subtraction of two regimes signifies an increase or decrease the successive 1/0s count. Therefore, instead of decoding R_A/R_B to decimal R_{DA}/R_{DB} to get $R_{DA} + R_{DB}$ and then encoding $R_{DA} + R_{DB}$ into regime format, *BRPU* decodes R_A to decimal R_{DA} to shift the R_B for R_{DA} bits to get R'_B and concatenates R_A with R'_B via 12 b bit wise or operation. It removes one regime's decoder and replaces the original adder/encoder with shifter/OR, saving 23.6% of area and 22.2% of power. During the shift-OR-based processing, a bidirectional shift controller schedules BRPU to decode the small $|R|$ while shifting large $|R|$. It reduces the shifting energy for R_A/R_B with the same sign since a smaller shift code indicates less shifting power. Besides, it avoids the shifting error for R_A/R_B with the different sign. Specifically, once the shift step is larger than the R 's bit-width, it moves all R away, introducing shifting errors. The bidirectional shift controller detects the small $|R|$ with 12 b XOR and a 12b leading-zero detector for R_A/R_B with the same sign based on the theory that less successive 0/1 indicate small $|R|$ and denotes early $0 \rightarrow 1$ or $1 \rightarrow 0$ change. It uses XOR and a leading-one detector if R_A/R_B with different signs. BRPU reduces the regime processing energy by 68%.

[Figure 34.1.4](#) depicts the CPCS, increasing the CIM array utilization. The CPCS CIM core consists of an 8×48 CIM array, a load controller, a critical-bit computing unit, and a 3:8 decoder. The 48b cell of each row is separated into 12 CIM units, each containing 4 CIM cells. The 4b CIM unit has 3 work modes: (1) when storing 3b $W[2:0]$, it receive $A[n] \mid A[n+2]$ in each cycle to get $P[2:0] = W \times (A[n] \mid A[n+2])$. A dual-bit MAC unit uses $P[2:0]$ to obtain $O = W \times A[n] + W \times A[n+2]$. If $A[n+2]/A[n] = 00$ or 01 then $O = P[2:0]$. If $A[n+2]/A[n] = 10$ then $O = P[2:0] \ll 2$. Otherwise, $O = P[2:0] + P[2:0] \ll 2 = [0, 0, P_2, P_1, P_0] + [P_2, P_1, P_0, 0, 0]$. The dual-bit MAC unit gets O by computing 3b $S[3:0] = [0, 0, P_2] + [P_2, P_1, P_0]$ and concatenates $[P_1, P_0]$. The computation of $S[1]$ consumes the largest number of transistors, defined as a critical bit, where $S[1] = W[P] = W[1] \wedge (W[0] \& W[2])$. Thus, rather than computing $S[1]$ at runtime, CPCS pre-computes and stores it in the spare cell. (2) When storing the 4b $W_n[3:0]$ with $W_n[0] = 0$, the CIM unit regards $W_n[3:1]$ as a 3b weight. It uses the spare bit from the $2b W_m[1:0]$ to store $W_n[P]$ since the $2b W_m[1:0]$ does not need extra $W_m[P]$ for dual-bit MAC. The 2-4b weight set-associate approach increases the throughput by 31%. (3) When storing the 4b $W_n[3:0]$ with $W_n[0] = 1$, it performs standard bit-serial computing. The load controller, critical-bit computing unit, and 3:8 decoder work during weight loading to compute $W[P]$ and associate 2-4b weights, whose overhead is only 1.6%. CASU increases 36.3% of CIM utilization.

[Figure 34.1.5](#) details the CASU, saving energy of the adder tree. In each cycle, CASU receives 2 exponents to detect whether their mantissa multiplication results, defined as $M_A = W_0 \times A$ and $M_B = W_1 \times B$, have overlap bits after alignment. M_A and M_B are the bit-serial computing results from the CIM array. Once there are no overlap bits, CASU performs $M_A \mid M_B$ to replace $M_A + M_B$. If M_A and M_B have overlap bits, instead of directly performing $M_A + M_B$, the scheduler permutes the bit-serial computing order of M_A and M_B to create no overlap conditions. For instance, if $M_A = W_0 \times A$ and $M_B = W_1 \times B$ have 1 overlap bit, it indicates that all of the bit-serially computed $W_0 \times A[i]$ and $W_1 \times B[i]$ have 1 overlap bit. Therefore, every $W_0 \times A[i] + W_1 \times B[i]$ requires an adder. Rather than synchronously computing $A[i]$ and $B[i]$ in each cycle, the scheduler cyclically shifts A by 1b, which permutes A 's bit-serial computing from $A[0:n]$ to $A[1:n] - A[0]$. The CIM array computes $A[i+1]$ and $B[i]$ together, where $W_0 \times A[i+1]$ and $W_1 \times B[i]$ have no overlap bit if $i < n$. The overlap only happens in $W_0 \times A[0] + W_1 \times B[n]$, which uses 1 adder. As a result, CASU saves n adders, instead using $(n-1)$ OR with 1 adder. This approach increases a 36.4% no overlap ratio for a 31.1% power reduction. CASU reduces 56.9% of the energy for the adder tree for POSIT (8,1).

[Figure 34.1.6](#) shows the measurement results. The DP-CIM processor consumes 5.5 - 237mW of energy with a voltage of 0.55 - 1.0V for 78 - 419MHz frequency. Its peak energy efficiency is 83.23TFLOPS/W for POSIT(8,1), which is achieved at 0.55V with a 78MHz, where 1 operation denotes 1 data wise multiplication or accumulation. Under the same accuracy, DP-CIM decreases the energy by 21% than [\[4\]](#) and $2.27\times$ than [\[5\]](#) for two reasons. First, BRPU avoids complicated codec logic of regime processing to save power. Second, CASU uses bit wise or operation to reduce energy of the adder tree. In addition, DP-CIM reaches a 49% speedup than [\[3\]](#) since CPCS can perform dual MAC in each cycle. When training on ResNet18, GPT-2, and ViT-B models, DP-CIM saves energy by $8.81\times$, $7.90\times$, and $7.12\times$, respectively. [Figure 34.1.7](#) shows a summary and die photo of the processor fabricated in 28nm CMOS technology with an area of 1.41mm².



Figure 34.1.1:
Dynamic property of POSIT introduces challenges for applying POSIT in CIM.

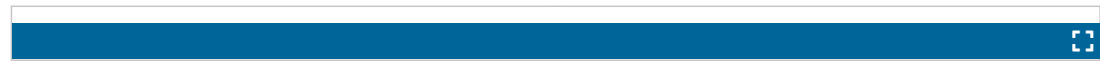


Figure 34.1.2:
The overall architecture of the proposed POSIT digital CIM (PD-CIM).



Figure 34.1.3:
Bi-directional regime processing unit (BRPU) replaces the codec with bi-directional shifting and an or gate by using the successive 0/1 coding property of regime.



Figure 34.1.4:
A critical-bit pre-compute-and-store (CPCS) CIM utilizes spare bits to pre-compute and store a logical value that facilitates the CIM to achieve intra-CIM dual-bit MAC computing.



Figure 34.1.5:
Cyclically alternating computing scheduling unit (CASU) uses bit-wise or operations to replace addition for no overlap mantissa and changes the computing order to increase more no overlap mantissa.



Figure 34.1.6:
Measurement results and performance comparison table.



Figure 34.1.7:
Chip micrograph and performance summary.

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